

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Design and Analysis of Algorithms

COURSE CODE: CSA06

COURSE OUTCOME COVERED

CO2: Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1,BL2,BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 10

Total Marks: 50

Annexure A

Scenario Based Assessment

Code of Conduct: I C .pavani/(192524260)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:C.PAVANI

Annexure A

Scenario-Based Assessment

QUESTION 16:

CODE:

```
#include <stdio.h>

int main() {
    int a[100], n, key, i;
    printf("Enter number of products: ");
    scanf("%d", &n);
    printf("Enter product IDs:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    printf("Enter product ID to search: ");
    scanf("%d",&key);
    for(i=0;i<n;i++) {
        if(a[i]==key) {
            printf("Product found at position %d", i+1);
```

```

        return 0;
    }
}

printf("Product not found");
return 0;
}

```

The screenshot shows the Programiz C Online Compiler interface. The code editor on the left contains the following C program:

```

11     scanf("%d",&a[i]);
12
13     printf("Enter product ID to search: ");
14     scanf("%d",&key);
15
16     for(i=0;i<n;i++) {
17         if(a[i]==key) {
18             printf("Product found at position %d", i+1);
19             return 0;
20         }
21     }
22
23     printf("Product not found");
24     return 0;
25 }

```

The output window on the right shows the following execution results:

```

Enter number of products: 5
Enter product IDs:
210
231
453
453
765
Enter product ID to search: 231
Product found at position 2
=== Code Execution Successful ===

```

QUESTION 17:

CODE:

```

#include <stdio.h>

int main() {
    int x[10], y[10], n, i, j;
    float d, max = 0;
    printf("Enter number of points: ");
    scanf("%d",&n);
    for(i=0;i<n;i++) {
        scanf("%d%d",&x[i],&y[i]);
    }
    for(i=0;i<n;i++) {
        for(j=i+1;j<n;j++) {
            d = (x[i]-x[j])*(x[i]-x[j]) + (y[i]-y[j])*(y[i]-y[j]);

```

```

        if(d > max)
            max = d;
    }
}

printf("Maximum squared distance = %.2f", max);
return 0;
}

```

The screenshot shows the Programiz Online C Compiler interface. The code editor on the left contains a C program for finding the farthest pair of points. The output window on the right shows the program's execution results.

```

main.c
23     for(j = i + 1; j < n; j++)
24     {
25         distance = sqrt((x[j] - x[i]) * (x[j] - x[i]) +
26                         (y[j] - y[i]) * (y[j] - y[i]));
27
28         if(distance > maxDistance)
29         {
30             maxDistance = distance;
31             p1 = i;
32             p2 = j;
33         }
34     }
35 }
36
37 printf("\nFarthest Pair of Points:\n");

```

Output:

```

Enter number of points: 4
Enter the coordinates (x y):
0 0
2 1
5 4
1 3

Farthest Pair of Points:
(0, 0) and (5, 4)
Maximum Distance = 6.40

=== Code Execution Successful ===

```

QUESTION 18:

CODE:

```

#include <stdio.h>

#include <string.h>

int main() {
    char text[100], pat[50];

    int i, j;

    printf("Enter text: ");

    gets(text);

    printf("Enter pattern: ");

    gets(pat);

```

```

for(i=0;i<=strlen(text)-strlen(pat);i++) {
    for(j=0;j<strlen(pat);j++) {
        if(text[i+j]!=pat[j])
            break;
    }
    if(j==strlen(pat)) {
        printf("Pattern found at position %d", i);
        return 0;
    }
}
printf("Pattern not found");
return 0;
}

```

The screenshot shows the Programiz C Online Compiler interface. The code editor on the left contains a C program that prompts the user to enter a text string and a pattern, then checks if the pattern is found in the text. The output window on the right shows the program's execution with the input 'computer science' and 'science', resulting in the message 'Pattern found at position 9'. The status bar at the bottom indicates 'Code Execution Successful'.

```

main.c
1 #include <stdio.h>
2 #include <string.h>
3
4 int main() {
5     char text[100], pat[50];
6     int i, j;
7
8     printf("Enter text: ");
9     fgets(text, sizeof(text), stdin);
10
11     // Remove newline character from text
12     text[strcspn(text, "\n")] = '\0';
13
14     printf("Enter pattern: ");
15     fgets(pat, sizeof(pat), stdin);

```

Output

```

Enter text: computer science
Enter pattern: science
Pattern found at position 9

=== Code Execution Successful ===

```

QUESTION 19:

CODE:

```

#include <stdio.h>

int main() {
    int a[100], n, key;
    int low=0, high, mid;
    printf("Enter number of salaries: ");
    scanf("%d",&n);
    printf("Enter sorted salaries:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&a[i]);

```

```

printf("Enter salary to search: ");
scanf("%d",&key);
high = n-1;
while(low<=high) {
    mid = (low+high)/2;
    if(a[mid]==key) {
        printf("Salary found at position %d", mid+1);
        return 0;
    }
    else if(key<a[mid])
        high=mid-1;
    else
        low=mid+1;
}
printf("Salary not found");
return 0;
}

```

The screenshot displays the Programiz Online C Compiler web interface. The browser's address bar shows the URL `programiz.com/c-programming/online-compiler/`. The interface includes a top navigation bar with the Programiz logo, a sidebar with various icons, and a main workspace divided into a code editor and an output console.

Code Editor (main.c):

```

25     printf("Salary found at position %d\n", mid + 1);
26     return 0;
27 }
28 else if(key < a[mid]) {
29     high = mid - 1;
30 }
31 else {
32     low = mid + 1;
33 }
34 }
35
36 printf("Salary not found\n");
37
38 return 0;
39 }

```

Output Console:

```

Enter number of salaries: 5
Enter sorted salaries:
10000
15000
20000
25000
30000
Enter salary to search: 25000
Salary found at position 4

=== Code Execution Successful ===

```

The bottom of the image shows a Windows taskbar with the Start button, a search bar, and several application icons. The system clock in the bottom right corner indicates the time is 13:37 on 01-08-2026.

QUESTION 20:**CODE:**

```
#include <stdio.h>

int main() {
    int x[10], y[10], n;
    int i, j, k, side;
    printf("Enter number of points: ");
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d%d",&x[i],&y[i]);
    printf("Hull edges are:\n");
    for(i=0;i<n;i++) {
        for(j=i+1;j<n;j++) {
            side = 0;
            for(k=0;k<n;k++) {
                int val = (x[j]-x[i])*(y[k]-y[i]) - (y[j]-y[i])*(x[k]-x[i]);
                if(val>0)
                    side++;
                else if(val<0)
                    side--;
            }
            if(side==n || side==-n)
                printf("(%d,%d) -> (%d,%d)\n", x[i], y[i], x[j], y[j]);
        }
    }
    return 0;
}
```

CO2 Assessment 2 Scenario Ba...Online C Compiler - Programiz

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programiz.com/c-programming/online-compiler/

Programiz

C Online Compiler

MIC

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main.c

Share

Run

Output

Clear

```
1 #include <stdio.h>
2
3 int main() {
4     int x[10], y[10], n;
5     int i, j, k, side;
6
7     printf("Enter number of points: ");
8     scanf("%d",&n);
9
10    for(i=0;i<n;i++)
11        scanf("%d%d",&x[i],&y[i]);
12
13    printf("Hull edges are:\n");
14
15    for(i=0;i<n;i++) {
```

```
Enter number of points: 4
0 0
0 4
4 4
4 0
Hull edges are:

=== Code Execution Successful ===
```

GoDaddy

Find Yours

Search

ENG IN

13:39 01-08-2026